

Summary

Embedded systems engineer with hands-on experience in Embedded C/C++, FreeRTOS, ARM64 assembly, MCU firmware, Lattice FPGA logic, serial interfaces, robotics, and engineering validation. Comfortable moving from low-level firmware through oscilloscope-based debugging, field testing, and C++/CUDA host-side systems tools.

Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed Embedded C and FreeRTOS sensor-logging and control functionality for geological well tools using Lattice FPGA, PIC32, dsPIC, ESP32, STM32, and C8051-class platforms.
- Supported board- and system-level bring-up with oscilloscope-based debugging, field validation, and targeted PCB updates in Altium.
- Built C++/CUDA parallel-computing applications for waveform processing and field diagnostics; created C++ GUI applications with OpenGL custom plotting controls for large sensor datasets.
- Supported those deployed tools with a C#/ASP.NET web server and Microsoft SQL Server for remote calibration and data collection.

Yale University

Embedded Systems Teaching Assistant / PhD Researcher

2021–present

- Supported embedded systems instruction and student lab work involving firmware, hardware interfaces, debugging, and instrumentation.
- Brought a strong embedded perspective into broader scientific-computing and engineering workflows during PhD research.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026, strengthening the research side of my hardware-software profile.

Selected Projects

EEG Robotic Hand: Built a PIC32-based robotic-hand controller driven by EEG sensor input, integrating biosignal acquisition, embedded control logic, and servo actuation.

Maze Mapping Robot: Built a PIC32 robot that mapped a maze and transmitted the map for remote visualization; used an FPGA programmed in Quartus to support mapping and display workflows.

MIPS Processor: Single-cycle Verilog CPU with self-checking test bench for readable digital-design validation.

Matrix-Multiply Accelerator: Small Verilog accelerator block showing datapath and control concepts relevant to embedded ML hardware.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Firmware: Embedded C/C++, FreeRTOS, ARM64 assembly, PIC32, dsPIC, ESP32, STM32, C8051, sensor interfaces

Hardware: Verilog, Lattice FPGA, Quartus, board bring-up, Altium PCB updates

Debug: Oscilloscopes, bench instrumentation, field testing, board bring-up

Systems Software: Python, C++, CUDA, OpenGL, Linux, C#; C#/ASP.NET and Microsoft SQL Server for internal calibration services

Serial Interfaces: SPI, I2C, UART/USART, serial device control, embedded logging workflows